Happy Hoppy Frogs
Development Office
Spring 2025

Element K: Reflection on the Design Project

The development office at our independent boarding and day school has an overflow of items. The excess items piled on the floor hinder workflow and efficiency. Many items move around the development office; therefore, it is important to keep them organized. The current storage situation is unsatisfactory in the eyes of school donors and visitors.

To define the problem, we interviewed primary stakeholders to find constraints and high-priority requirements. We wrote a letter to Ms. C and Ms. P and received feedback. Boxes will be concealed to give the office a nice and clean look. Our group did a good job to define the stakeholder's problem. We got good feedback from the stakeholders when we shared our design idea.

For brainstorming, each team member created 20 designs on paper. We then discussed as a group to find the best idea for the desk. We found that making ideas is better when written out. We agreed to make multiple designs and not one detailed design. While brainstorming, it would have been best if we had created more designs.

Next, the top five designs were selected. We made a weighted decision matrix that accounted for major design requirements and scored each idea for the design requirements. After each design was scored, we chose the design with the most points and improved it until it was effective. This worked well because it helped us decide on what to do for our desk and fit our stakeholder needs.

To prototype the design, we used OnShape to create our designs. OnShape was easy to make changes to our design. This was cheaper and less time consuming than making a physical prototype.

In testing and evaluation, we didn't build a physical prototype due to time constraints, but our maintenance team will build it over the summer. If we were to have built a physical prototype, we would need the shelves sturdy enough to hold heavy boxes. We would make sure shelves had a back on them as well. We would paint the shelves white, which would help them match the appearance of the room. At the same time, we need to design and build another cabinet as required. It needs to have a lock and a door. It would have been best if we were able to test our design, but due to time constraints this was not possible.

The first iteration was a brief notebook sketch of a folding shelf, the shelf was removed. The prototype was made in Onshape with more shelves attached together. We then met with the stakeholders, and created our second prototype. This prototype was evaluated with experts, and additions to material, color, and dimensions were made. The third prototype also underwent the last modification. The data was changed, all the lengths, and the thickness of wood from 1 inch to 3/4 in. We used a CAD software called Onshape to create our designs. Onshape allowed us to see what it would look like in real life, making it easier to build and more cost-effective. It also allowed us to change our design. However, it didn’t allow us to see if our design would function. While it was cost-effective, it also required us to build our prototype in Onshape, which could be very time-consuming. Building a model is not easy if you only do rapid prototyping or digital prototyping; if both styles are used, it creates better and efficient prototyping.

Our largest constraint was time, and how we used that time. It would have been more efficient if everyone had been prepared. Our group does not know how to build cabinets, this is another problem that we faced. Our group used OnShape to create our prototype. This was a useful tool, but prevented us from having a physical prototype. We also learned how to brainstorm and work together as a team during this process.